THE PRESIDENT'S OFFICE

REGIONAL ADMINISTRATION AND LOCAL GOVEVERNMENT

FORM TWO MOCK EXAMINATION 2018

LONGIDO DISTRICT SYNDICATE

BIOLOGY

Time 2:30 Hours

Instructions

1. This paper consists of sections A,B and C with a total of **eleven(11)** questions.
2. Answer all questions in section A and B,and **one(1)** question from section C.
3. All writing must be in blue or Black ink **except** drawing which must be in pencil.
4. Calculators and cellular phone are not allowed in the examination room.
5. Write your **Examination Number** at the top right corner of every page.

SECTION A (30 Marks)

1. For each item(i)-(x), choose the correct answer among the given alternatives and write its letter beside the item number in the space provided

i. The purpose of control in an experiment is to
   A. Identify the problem
   B. Test the hypothesis
   C. Display results
   D. Confirm conclusion
   E. Predict the results

ii. Incineration is the best method for burning
   A. Gaseous waste
   B. Hazardous waste
   C. Non organic solid waste
   D. Liquid waste
   E. All types of waste

iii. In the microscope the eye piece lens reads x10 and objective lens reads x40, the total magnification.
   A. X400
   B. X200
   C. X300
   D. X4000
   E. X20

iv. Oxygen dept occurs when
   A. Too much carbon dioxide is present in body
   B. The rate of respiration exceed that photosynthesis in green plant
   C. Alcohol is formed in tissue

D. There is insufficient oxygen in muscles tissue during the strenuous exercise
E. There is insufficient lactic acid in muscle tissue

v. Which of the following sets of enzymes is responsible for protein digestion?
A. Amylase and maltase
B. Amylase and rennin
C. Maltase and pepsin
D. Maltase and lipase
E. Pepsin and rennin

vi. Amoeba, plasmodium, Euglena and Paramecium belong to kingdom
A. Protoctista
B. Monera
C. Fungi
D. Animalia
E. None of them

vii. In which of the following mammalian blood vessels does deoxygenated blood flow away from the heart
A. Aorta
B. Vena cava
C. Pulmonary artery
D. Pulmonary vein
E. None of the above

viii. The concept of good health implies……..health
A. Sexual, physical and mental
B. Physical, mental and social
C. Mental, sexual and physical
D. Physical, mental and family
E. Reproductive, social and family

ix. In biology Human being, Lizard, Fish and Elephant belong to which branch of biology?
A. Botany
B. Anemology
C. Zoology
D. Mycology
E. Virology

x. Ways of treating food to make it edible, appetizing and safe to it or to keep fresh for a long time is referred to as
A. Food preservation
B. Food processing
C. Food storage
D. Food manipulation
E. Balance diet

2. For each of item(i)-(ix),write **TRUE** if a statement is correct **or FALSE** if a statement is not correct in a space provided
   i. Removal of pancreas from an animal will result failure to digest protein in small intestine_____
   ii. Some laboratory accident cannot be avoided by reading warning signs on a chemical container_____
   iii. The sign of human skull and bones on a chemical container indicate the chemical in that container is explosive_____
   iv. Erythrocyte are blood cell which defend the body against pathogens_____
   v. Phloem tissue are responsible for transport water in vascular plant_____
   vi. Hypertension can be prevented by engaging in regular exercise and avoid alcoholism_____
   vii. Lymph is formed from tissue that does not flow back into the capillaries_____
   viii. Mosses and Liverworts have true roots ,stem and leaves_____
   ix. Heartburn is a disorder of digestive system_____
   x. Individuals with blood group AB are called Universal Recipient_____

3. Match the phrase in LIST A with responses in LIST B by writing the latter of correct responses from LIST B below the item number of LIST A in a table provided

LIST A	LIST B
i. The first part of small intestine	A. Lipase
ii. Control the flow of material out of the stomach	B. Erepsin
iii. Wave like muscular contraction push food along the oesophagus	C. Duodenum
	D. Colon
iv. An enzyme that converts lipids to fatty acid and glycerol	E. Cardiac sphincter
v. It contain acidic medium for digestion	F. Pyloric sphincter
	G. Stomach
	H. Liver
	I. Digestion
	J. Peristalsis

4. Complete each of the following statements by writing the correct answer in the answer sheet provided
   i. A valve between the left atrium and left ventricle of the heart is called_____
   ii. Large vein that empties into the left atrium of the heart is_____
   iii. The tube that form branching system and carry blood to the heart is_____
   iv. Hardening and narrowing of an arteries is_____
   v. A disorder in which high blood pressure is the primary symptoms is_____

SECTION B (50 Marks)

5. a)(i) What do you understand by the term laboratory?

   (ii) Outline four qualities of a good laboratory

b) Mention one function of each of the following apparatus

(i) Mortar and pestle

(ii) Petri dish

(iii) Hand lens

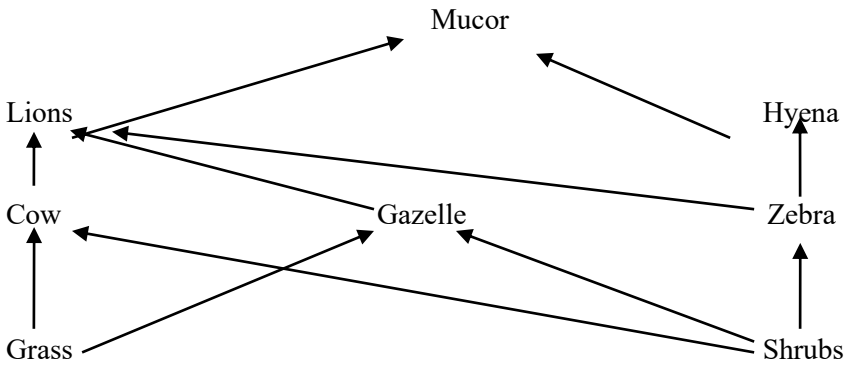
(iv) Thermometer

6. (a) Explain the mechanism of gaseous exchange in mammals

(b) Give out respiratory surfaces of the following organisms

i.	Amoeba	
ii.	Insects	
iii.	Spider	
iv.	Fish	
v.	Plants	
vi.	Amphibians	
vii.	Mammals	
viii.	Birds	
ix.	Reptiles	

7. Study the pattern bellow and answer all questions that follows



a)(i) What is the name of feeding relationship represented above?

(ii) What does sequence the sequence represent?

Shrubs → Zebra → Hyena → Mucor

b) Name the organisms which are

(i) Secondary consumers

(ii) Primary producers

c) What is the role of Mucor in the pattern above?

8. a)(i) What is tissue

(ii) Name three animal and two plant tissue

b) State three structure difference between cell wall and cell membrane

9. (a) Define the term Photosynthesis

(b) List down three point to show the economic importance of photosynthesis

(c) With the aid of well balanced equation of photosynthesis, mention raw materials and end products of photosynthesis

SECTIONS C (20 Marks)

Answer **one (1) question** from this section

10. Describe the causes, mode of transmission and prevention measures of Tuberculosis disease in the respiratory system
11. Biology is not an independent field of study but it is related to many other fields. Explain the validity of this statement